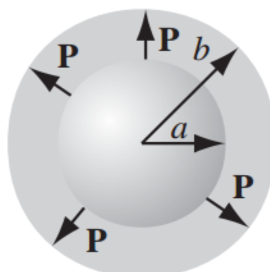
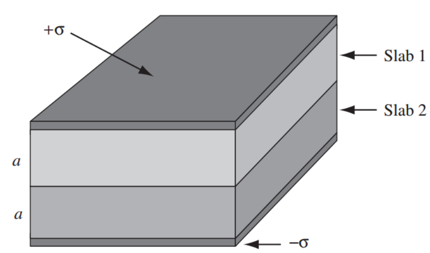
Indian Institute of Technology Guwahati
Department of Physics
PH102
Tutorial-6

Date: 20-02-2024

1. A thick spherical shell (inner radius a, outer radius b) is made of dielectric material with a "frozen-in" polarization $\vec{P}(\vec{r}) = \dfrac{k}{r}\hat{r}$, where k is a constant and r is the distance from the centre. Find $\vec{D}$ and $\vec{E}$ in all three regions. Note that there is no free charge in the problem.



2. The space between the plates of a parallel-plate capacitor is filled with two slabs of linear dielectric material. Each slab has thickness a, so the total distance between the plates is $2a$. Slab 1 has a dielectric constant of 2, and slab 2 has a dielectric constant of 1.5. The free charge density on the top plate is σ and on the bottom plate $-\sigma$.



(a) Find the electric displacement $\vec{D}$ in each slab.

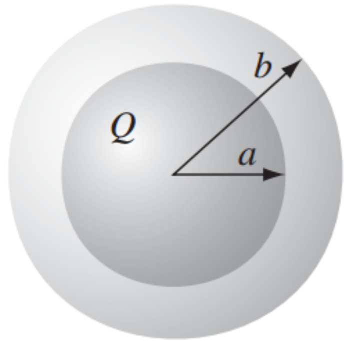
(b) Find the electric field $\vec{E}$ in each slab.

(c) Find the polarization $\vec{P}$ in each slab.

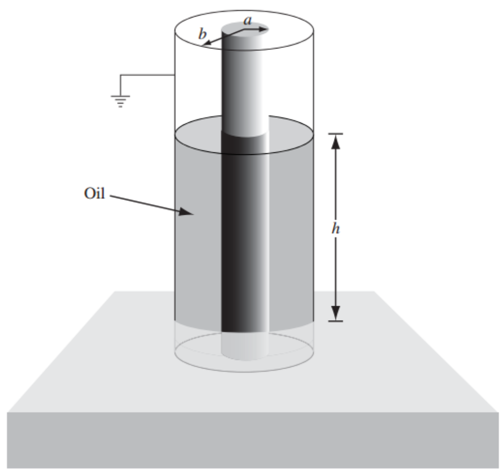
(d) Find the potential difference between the plates.

(e) Find the location and amount of all bound charge.

3. A spherical conductor of radius a carries a charge Q. It is surrounded by a linear dielectric material of susceptibility χ_e, out to radius b. Find the energy of this configuration.



4. Two long coaxial cylindrical metal tubes (inner radius a, outer radius b), stand vertically in a tank of dielectric oil (susceptibility χ_e, mass density ρ). The inner one is maintained at potential V, and the outer one is grounded. To what height h does the oil rise in the space between the tubes? Assume that the force experienced by the oil column is given by
$$F = \frac{1}{2} V^2 \frac{dC}{dh}$$
where C is the capacitance of the given system.



5. Two homogeneous dielectric regions 1 ($s \leq 4cm$) and 2 ($s \geq 4cm$) have dielectric constants 3.5 and 1.5, respectively. If $\vec{D_2} = 12\hat{s} - 6\hat{\phi} + 9\hat{z}$; calculate
   (a) $\vec{E_1}$ and $\vec{D_1}$,
   (b) $\vec{P_2}$ and ρ_2,
   (c) the energy density for each region.